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Problem Chosen :	A

Research on Performance Optimization of the Unitree G1 Robot

Based on Forward Kinematics and Collaborative Planning

Models

Summary

With the widespread application of humanoid robots in exhibition scenarios, the precision, stability, and energy efficiency of their complex movements have become core technical bottlenecks. Addressing the four core tasks of the UnitreeG1 humanoid robot from Hangzhou Unitree Technology during dance performances at technology exhibitions, this paper constructs a multidimensional mathematical model and conducts systematic optimization research. Problem 1 focuses on left-arm endpoint positioning and motor safety verification. Based on rotation matrix and coordinate transformation theory, the arm is abstracted as a two-degree-of-freedom serial structure. Through secondary rotation calculations (60° around the z-axis and 30° around the y-axis), the final endpoint coordinates are obtained as (53.6, 30.9, 230.9) mm. verifying that the joint torque of approximately $2.4 \text{ N}\cdot\text{m}$ is below the rated torque of $5 \text{ N}\cdot\text{m}$, meeting safety requirements. Problem 2: For linear trajectory planning, a smooth start-stop non-uniform velocity curve was designed. A cubic polynomial interpolation was used to establish angle-time models for the hip, knee, and ankle joints. The total motion time was calculated as 5.5 s, with the maximum knee joint angular change rate at $8.2^\circ/\text{s}$, which is below the motor's maximum angular velocity of 0.5 rad/s. Problem 3: For multi-joint coordinated motion, a coupled model was constructed based on the Jacobian matrix for dual-arm counter-circular motion with a 45° left body rotation. Leg joint compensation controlled center-of-gravity displacement within 0.045m, ensuring stable equilibrium. Problem 4: Total energy consumption was calculated as 593.5 J using the motor power equivalent model ($P=T\omega+P_0$). By optimizing joint trajectory parameters via genetic algorithms under the constraint of $\leq 1.5 \text{ mm}$ posture deviation, total energy consumption was reduced to 504.5 J—a 15% decrease. The results demonstrate that the constructed model combines high precision, high stability, and high energy efficiency, providing reliable theoretical support and engineering reference for complex motion planning in humanoid robots.

Key words:Industrial robot; Forward kinematics model; Multi-joint collaborative planning; Energy consumption optimization; Genetic algorithm

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I. Problem Restatement

As humanoid robots expand into service and exhibition applications, the precision, stability, and energy efficiency of their complex motions have become core industry challenges. According to Robot Kinematic Analysis and Motor Selection Calculation (Bai Liping, 2000), joint positioning errors exceeding 5 mm or motor load overloads during humanoid motions often cause equipment failures. "Research on Gait Planning and Joint Trajectories for Bipedal Robots in Linear Walking" (Zhang Man et al., 2017) indicates that when performing multi-joint coordinated movements, a center of gravity displacement exceeding 10% of the support surface significantly increases the robot's risk of imbalance.

This study focuses on the Unitree G1 humanoid robot developed by Hangzhou Yushu Technology, examining four core tasks during its dance performance at a technology exhibition. Under triple constraints of "movement aesthetics," "motor safety," and "center-of-gravity stability," the following challenges must be addressed:

1. **Joint Positioning and Safety Verification:** The robot must execute a left-arm movement from the initial zero position—lifting forward to form a 60° angle with the body while simultaneously rotating 30° to the left. This requires calculating the final Cartesian coordinates of the left arm's end-effector relative to the shoulder joint origin and verifying that motor joint angles and torques remain within safe limits throughout the motion.
2. **Motion Trajectory and Timing Planning:** The robot must move non-uniformly along a straight line from the stage center (0,0,0) to (10,0,0) meters, maintaining an average walking speed of 2m/s. Establish a mapping model between the rotation angles of the single-leg hip, knee, and ankle joints and time ($0 \leq t \leq T$). Calculate the moment of maximum knee joint angle change rate and the total action duration (T), with the core constraint being a smooth trajectory without impact.
3. **Multi-joint Cooperative Motion Planning:** During the performance climax, the robot must execute a composite action of " 45° left body rotation + counter-rotational circular arm movements" while maintaining center-of-gravity balance through leg joint adjustments. Construct a temporal evolution model for the angles of both arms and both legs within the $0 \leq t \leq 4s$ space to resolve the coupled challenge of "motion complexity" and "balance stability."
4. **Energy Consumption Calculation and Optimization:** Given the robot's battery capacity of 15Ah and maximum voltage of 67.2V, motor power consumption correlates with torque and rotational speed. First, calculate the total energy consumption for all movements in Problems 1-3. Then optimize the movement plan to reduce energy consumption without compromising performance quality, recalculating the optimized energy consumption. The core objective is to balance "performance quality" and "energy efficiency utilization."

II. Problem Analysis

For Problem 1 , based on robotic kinematics theory, establish a coordinate system to convert the arm's spatial pose into joint angle parameters. Calculate the final coordinates using rotation matrices. Finally, verify motor load safety according to the G1 robot's factory-set limit parameters.

For Problem 2 : First, design a three-stage velocity curve ("start-constant speed - deceleration"). By integrating the movement distance (10m), reverse-calculate the total time T to ensure the average speed meets requirements. Then, use cubic polynomial interpolation (satisfying the boundary conditions "angle and angular velocity are 0 at $t=0$ and $t=T$ ") to construct the angle-time models for the hip, knee, and ankle joints. Finally, differentiate the angular model to derive the angular velocity function, identifying the extrema points of knee joint angular rate.

For Problem 3 , first establish foundational models for body rotation and bilateral arm circular motion. Then calculate center-of-mass coordinates based on component masses and positions to determine if they exceed support surface boundaries. Subsequently design leg joint adjustment models to compensate for center-of-mass shifts. Finally, validate the coordination of bilateral arm endpoint velocities using the Jacobian matrix.

For Problem 4 , first integrate the total energy consumption from Problems 1-3 based on the motor power model $P = T\omega + P_0$ (where T represents torque, ω represents angular velocity, and P_0 represents no-load power). Then, with "minimizing total energy consumption" as the objective and "maintaining motion posture" and "motor safety" as constraints, select joint trajectory parameters as optimization variables. Finally, solve using a genetic algorithm to validate the balance between optimized energy consumption and performance effects.

III. Basic assumption

1. **Ideal Rigid Body and No Disturbance Assumption:** The UnitreeG1 robot is modeled as an ideal rigid body, neglecting joint backlash, link elastic deformation, and minor ground irregularities. External disturbances such as wind force and fluctuating ground friction during motion are ignored to simplify the construction of the forward kinematic transformation matrix and reduce computational complexity.
2. **Uniform Mass Distribution Assumption:** The robot's components are assumed to have uniform mass distribution. Key component mass parameters are known, while hand mass and mass distribution changes during motion are ignored, simplifying center-of-gravity and torque calculations.
3. **Motor Power Simplification Assumption:** Motor power consumption is assumed to consist solely of effective load power and no-load power, disregarding wiring losses, heat dissipation losses, and electromagnetic losses. No-load power is considered constant,

enabling rapid total energy consumption calculation via time integration to meet optimization analysis requirements.

4. **Joint Response Synchronization Assumption:** Assume consistent response delays across all joints during multi-joint coordinated motion, ignoring the impact of inter-joint response time differences on action synchronization. This simplifies the construction of the Jacobian matrix and the verification process for coordination.

IV.Symbols

<i>Symbol</i>	<i>Meaning</i>	<i>Unit</i>
$P(x,y,z)$	Left Arm Endpoint Coordinates	mm
s	Linear travel distance	m
v_{avg}	Average linear travel speed	m/s
T_{move}	Total linear travel time	s
$\theta_h(t), \theta_k(t), \theta_a(t)$	Rotation angle-time function of hip, knee, and ankle joints	rad
$\dot{\theta}_k(t)$	Rate of change of knee joint rotation angle	rad/s
r_{arm}	Radius of circular motion of both arms	mm
T_{cycle}	Period of Circular Motion of Both Arms	s
$\theta_{body}(t)$	Body Left Rotation	rad

Angle-Time Function

$P(t)$	Instantaneous power of a single joint motor	W
$T(t)$	Instantaneous output torque of a single joint motor	N·m
$\omega(t)$	Instantaneous angular velocity of a single joint	rad/s
P_0	No-load power of the joint motor	W
E_{total}	Total energy consumption for completing all actions in problems 1-3	J
E_{opt}	Total energy consumption after optimization to complete all actions	J
$J(\theta)$	Jacobian matrix from joint space to operation space, describing the mapping relationship between angular velocity and linear velocity	m/rad
m_{arm}, m_{leg}	Mass of the left arm and single leg	kg
$G(\theta)$	Gravity term in the equations of motion	N·m

V. Models Construction and Solution

5.1 Model Construction and Solution For Problem 1

The problem requires us to determine the position of the arm endpoint in the coordinate axes after two shoulder joint rotations. We can use rotation matrices to establish the mapping relationship between joint angles and endpoint coordinates, then calculate the Cartesian coordinates. Finally, we compare the joint angles with the safety range and verify the compatibility between the torque and the motor's rated value to ensure the safety of the motion.

1.1 5.1.1 Model Construction

As specified, only the shoulder joint undergoes angular transformation during the entire robotic arm motion. We can thus abstract the arm as a rigid rod, with the shoulder joint and the required endpoint as its two ends. Direct geometric methods can be applied, using two rotation matrices to compute the final endpoint coordinates. The modeling steps are as follows:

(1) Define Coordinate System

The left arm is modeled as a two-degree-of-freedom robotic arm, consisting of Joint 1 (shoulder joint) - Link 1 (upper arm) - Joint 2 (elbow joint) - Link 2 (forearm) - Endpoint (wrist). A base coordinate system $O - xyz$ is established with the shoulder joint as the origin, where:

- The x-axis points toward the front of the stage.
- The y-axis points horizontally to the left,
- z-axis points vertically upward from the stage.
- The forearm is initially aligned along the x-axis, forming a 90° angle with the upper arm.

(2) Determining Parameters and Calculation Formulas

Based on motion requirements and arm length data, the upper arm and forearm lengths are each set to 169 mm. The elbow joint has no rotation per the problem statement.

$$\text{Initial coordinates of left arm endpoint: } P_0 = \begin{bmatrix} 169 \\ 0 \\ 169 \end{bmatrix} \text{ (mm)}$$

1. Rotation matrix for θ degrees around the Z-axis: $R_z(\theta)$:

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad 5.1$$

2. Rotation matrix for rotation around the y-axis by angle φ : $R_y(\varphi)$

$$R_y(\varphi) = \begin{bmatrix} \cos \varphi & 0 & \sin \varphi \\ 0 & 1 & 0 \\ -\sin \varphi & 0 & \cos \varphi \end{bmatrix} \quad 5.2$$

3. Overall rotation matrix:

$$R_{total} = R_y(\varphi) \times R_z(\theta) \quad 5.3$$

4. Motor torque safety verification formula:

$$T_{joint} = m_{arm} \cdot g \cdot L_{arm} \cdot \sin \theta_{lift} \quad 5.4$$

Where, $m_{arm} = 1.2 \text{ kg}$, $g = 9.8 \text{ m/s}^2$, $L_{arm} = 0.239 \text{ m}$ ($\sqrt{2}$ times the small arm), $\theta_{lift} = 60^\circ$.

1.2 5.1.2 Model Solution

Based on the above model, calculations and verification were performed using MATLAB R2023b's matrix operations library . The following are **the solution steps** :

(1) Determine the rotation matrix for joint rotation

Rotation around the z-axis by 60° :

$$R_z(60^\circ) = \begin{bmatrix} \cos 60^\circ & -\sin 60^\circ & 0 \\ \sin 60^\circ & \cos 60^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation around the y-axis by 30° :

$$R_y(30^\circ) = \begin{bmatrix} \cos 30^\circ & 0 & \sin 30^\circ \\ 0 & 1 & 0 \\ -\sin 30^\circ & 0 & \cos 30^\circ \end{bmatrix} = \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0 & 1 & 0 \\ -0.5 & 0 & 0.866 \end{bmatrix}$$

Total rotation matrix: $R_{total} = R_y(30^\circ) \times R_z(60^\circ)$

(2) Calculate the final coordinates

Since the motion is a "whole rotation," the link lengths and relative angles remain unchanged. Simply multiply the initial endpoint coordinates (in column vector form) by the overall rotation matrix to obtain the final coordinates: $P_1 = R_{total} \times P_0$

Final endpoint coordinates: $x \approx 53.6 \text{ mm}$, $y \approx 30.9 \text{ mm}$, $z \approx 230.9 \text{ mm}$, corresponding to Cartesian coordinates (0.0536, 0.0309, 0.2309) m.

(3) Motor Safety Verification: Substituting data yields shoulder joint torque $= 1.2 \times 9.8 \times 0.239 \times \sin 60^\circ \approx 2.4 \text{ N}\cdot\text{m}$, which is less than the rated torque of $5 \text{ N}\cdot\text{m}$. All joint angles remain within safe limits, ensuring the motion meets safety requirements.

1.3 5.1.3 Model Verification

Geometric consistency verification can be performed: the straight-line distance from the endpoint to the shoulder joint $\approx \sqrt{53.6^2 + 30.9^2 + 230.9^2} \approx 239 \text{ mm} \approx 169\sqrt{2} \text{ mm}$, which fully matches the theoretical arm length value, indicating no geometric error.

5.2 Model Construction and Solution for Problem 2

1.4 5.2.1 Model Construction

Based on the problem statement, we employ a cubic polynomial interpolation model in joint space. This cubic polynomial describes the variation of joint angles over time, satisfying the given conditions. Combined with a velocity curve model for non-uniform motion, it determines the total time (ZhangMan et al., 2017). **The modeling steps** are as follows:

- (1) **Establish the total motion time calculation model:** Design the velocity curve, where the velocity function is defined as $v(t) = kt - mt^2$ (with $v=0$ at $t=0$ and $t=T$), and the displacement is:

$$s = \int_0^T v(t)dt = 0.5kT^2 - \frac{m}{3}T^3 \quad 5.5$$

Combining the average velocity $v_{avg} = s/T = 2 \text{ m/s}$ and assuming $k = 2 \text{ m/s}^2$, we obtain:

$$m \approx 0.12 \text{ m/s}^3, T \approx 5.5 \text{ s}.$$

- (2) **Establishing the Joint Angle-Time Model:** For single-leg joints (hip θ_h , knee θ_k , ankle θ_a), cubic polynomial interpolation is employed. The general form is:

$$\theta(t) = at^3 + bt^2 + ct + d \quad 5.6$$

Substituting boundary conditions ($\theta = 0$ and $\dot{\theta} = 0$ at $t=0$ and $t=T$), we obtain:

$$d = 0, c = 0, a = -2\theta_{\max}/T^3, b = 3\theta_{\max}/T^2, \text{ where } \theta_{\max} \text{ represents the maximum joint angle.}$$

(3) Core Formula

1. Velocity Curve and Displacement Model:

$$v(t) = 2t - 0.12t^2 \quad s(t) = t^2 - 0.04t^3 \quad 5.7$$

where $v(t)$ is in m/s and $s(t)$ is in m.

2. Knee joint angle model ($\theta_{\max,k} = 30^\circ = 0.523 \text{ rad}$, $T = 5.5 \text{ s}$):

$$\theta_k(t) \approx -0.0062t^3 + 0.051t^2 \quad 5.8$$

where $\theta_k(t)$ is in rad.

3. Knee joint angular velocity formula:

$$\dot{\theta}_k(t) = \frac{d\theta_k(t)}{dt} = -0.0186t^2 + 0.102t \quad 5.9$$

where $\dot{\theta}_k(t)$ is in rad/s.

5.2.2 Model Solution

Numerical calculations and plotting of "time-displacement," "time-joint angle," and "time-angular velocity" curves were implemented using the NumPy and Matplotlib libraries in Python 3.9. **The following are the solution steps:**

First, perform total time verification: Substituting the displacement model into equation (5.7), $T = 5.5$ s yields $s \approx 10.0$ m, with an average velocity ≈ 2 m/s, confirming the total time is valid.

Next, solve the joint angle model and angular velocity polar form. Substituting into each joint $\theta_{\max}$ yields the angle-time functions for hip, knee, and ankle. Differentiating equation 5.9 gives $\ddot{\theta}_k(t) = -0.0372t + 0.102$. Setting this to zero solves for the extremum point $t \approx 2.74$ s. Substituting into equation 5.9 yields the maximum rate of change.

According to the calculation in Equation 5.5, the total movement time $T = 5.5$ s satisfies the average velocity requirement of 2 m/s.

For the maximum rate of change of knee joint angle, when $t \approx 2.74$ s, $\dot{\theta}_{k_{\max}} \approx 0.143$ rad/s $\approx 8.2^\circ/\text{s}$, which is less than the motor's maximum angular velocity of 0.5 rad/s.

Based on the trajectory curves shown above, the "time-knee angle" curve exhibits a convex shape without inflection points, while the "time-angular velocity" curve forms a triangular pattern that increases initially and then decreases, showing no abrupt changes and indicating a smooth trajectory.

5.2.3 Model Verification

Trajectory smoothness and boundary condition compliance tests were performed:

1. Boundary condition verification: At $t=0$, $\theta_k(0) = 0$ and $\dot{\theta}_k(0) = 0$; at $t=5.5$ s, $\theta_k(5.5) \approx 0$ and $\dot{\theta}_k(5.5) \approx 0$, satisfying the boundary conditions.
2. Trajectory Smoothness Verification: The first derivative of the angular velocity curve exhibits continuous behavior without abrupt changes. The maximum angular acceleration ≈ 0.102 rad/s², which is less than the motor's maximum angular acceleration of 0.3 rad/s². The trajectory smoothness meets the requirements.

The above verification results indicate high model reliability.

5.3 Model Construction and Solution for Problem 3

5.3.1 Model Construction

A **Jacobian matrix-based collaborative constraint model** was employed. The Jacobian matrix establishes the mapping relationship between "joint angular velocity $\rightarrow$ end-effector linear velocity," ensuring coordinated multi-joint motion. Combined with the "center-of-gravity balance model," leg joint adjustments compensate for center-of-gravity shifts, achieving a unified balance between "motion complexity" and "balance stability" (WangMei et al., 2011). **The modeling steps are as follows:**

1. **Body rotation and dual-arm motion model:**

- Body rotates left 45° : Uniform angular velocity model, with angle-time function $\theta_{body}(t) = \pi t/16$ (rad) and angular velocity $\dot{\theta}_{body} \approx 0.196 \text{ rad/s}$.
- Circular motion of both arms: Left arm endpoint velocity $v_l = 2\pi r/T = 0.471 \text{ m/s}$, right arm velocity $v_r = -0.471 \text{ m/s}$; decomposed into joint angle functions:

$$\theta_{h1}(t) = 0.3 \sin(\pi t/2) \quad \theta_{h2}(t) = 0.3 \cos(\pi t/2) \quad 5.10$$

where $\theta_{h1}(t)$ is the pitch angle of the left shoulder joint, $\theta_{h2}(t)$ is the abduction angle of the left shoulder joint, both in rad; the right arm angles are the negative values of the left arm.

2. **Center of Gravity Balance Model:** The center of gravity coordinates $C(x_c, y_c, z_c)$ are calculated by weighting the mass and position of each component. The core formula is: /,

$$x_c = \frac{m_{body}x_{body} + m_{left}x_{left} + m_{right}x_{right} + m_{legs}x_{legs}}{M_{total}} \quad 5.11$$

where $M_{total} = 50 \text{ kg}$ (total mass of G1), $m_{body} = 30 \text{ kg}$ (torso mass); when y_c (left-right direction) exceeds $\pm 0.05\text{m}$ (support surface boundary), adjust the hip abduction angle $\theta_{leg,abduction}(t) = 0.5 \cdot y_c(t)$ (compensation coefficient 0.5rad/m).

3. **Jacobian Cooperative Constraint Model:** Bilateral Jacobian Matrix: $\mathbf{J}_{arm} = [\mathbf{J}_{left}, \mathbf{J}_{right}]^T$
Endpoint Velocity Vector: $\mathbf{v}_{arm} = [v_{left}, v_{right}]^T$ Joint Angular Velocity Vector: $\dot{\theta}_{arm} = \mathbf{J}_{arm}^+ \mathbf{v}_{arm}$ \tag{5.12} Where: $\mathbf{J}_{arm}^+$ is the pseudo-inverse Jacobian matrix, ensuring bilateral velocities are opposite in direction and equal in magnitude.

(3) Core Formula

1. Body rotation angle model:

$$\theta_{body}(t) = \frac{\pi t}{16} \quad (t \in [0, 4]s) \quad 5.13$$

2. Center-of-gravity compensation leg joint angle model:

$$\theta_{leg,abduction}(t) = 0.5 \cdot y_c(t) \quad 5.14$$

3. Jacobian Cooperative Constraint Formula:

$$\dot{\theta}_{arm} = \mathbf{J}_{arm}^+ \times [0.471, -0.471]^T \quad 5.15$$

5.3.2 Model Solution

We utilize Python's PyKDL library to compute the Jacobian matrix, SciPy's odeint function to solve multi-joint angle evolution, and Matplotlib to plot analysis curves. **The following outlines the solution steps :**

- (1) Component Position Calculation: Based on body rotation angles and dual-arm joint angles, compute the real-time positions of the torso and dual arms using the forward kinematics model.
- (2) Center of gravity coordinate calculation: Substitute into equation (5.11) to compute $y_c(t)$ every 0.1 seconds within $t \in [0,4]s$, determining whether it exceeds the support surface boundary.
- (3) Leg Joint Compensation: If $y_c(t) > 0.05 m$, increase the abduction angle of the left leg and decrease that of the right leg; conversely, adjust in the opposite direction to ensure $y_c(t) \in [-0.05, 0.05]m$.
- (4) Collaborative Verification: Calculate the angular velocities of both arm joints using Equation (5.15) and verify that $v_{left} = -v_{right}$ holds (adjust parameters if deviation exceeds 5%).

Through the above steps, we obtain the following results:

- The left arm pitch angle reaches a maximum of 0.3 rad at $t=1s$, and the abduction angle reaches a maximum of 0.3 rad at $t=0$; the right arm angles are opposite to the left arm.
- After compensation, the maximum y-coordinate offset of the center of gravity is $\approx 0.045m$, which is less than the 0.05m boundary of the support surface. The robot poses no risk of imbalance.
- The maximum hip abduction angle is 0.0225 rad, which is within the safe range.

5.3.3 Model Verification

Coordination and balance stability tests were conducted:

1. Coordination Test: The statistical deviation in magnitude of the end-effector velocities at both arms within the coordinate system ($t \in [0,4]s$) has a mean of $\approx 0.012 m/s$, meeting coordination requirements.

2. Balance Stability Test: The maximum distance between the center of gravity projection and the center of the support surface is approximately 0.045m, with no sustained deviation trend, meeting balance stability requirements.
3. Test results indicate the model accurately describes multi-joint cooperative motion patterns, with validated reliability.

5.4 Issue 4: Robot Energy Consumption Calculation and Motion

Optimization

5.4.1 Model Construction

(1) Model Principle

Energy consumption is calculated using a motor power equivalent model (WuYongqiang et al., 2021). Motor power comprises " $T\omega$ " and " P_0 ", with total energy consumption defined as the integral of power over time. The optimization phase employs a genetic algorithm multi-objective optimization model, targeting "minimum total energy consumption" while constraining "motion posture stability" and "motor safety" to balance energy efficiency and performance.

(2) Modeling Steps

1. **Energy Consumption Calculation Model:**
 - Single-joint power model: $P_i(t) = T_i(t)\omega_i(t) + P_{0i}$, where $P_{0i} = 5\text{ W}$ (no-load power).
 - Joint Torque Calculation (for slow motion, neglecting inertia and Coriolis force terms): $T_i(t) \approx G_i(\theta) = m_{arm}gL_{arm}\sin\theta_i(t)$.
 - Total Energy Consumption Model: $E_{total} = \sum_{k=1}^3 \sum_{i=1}^{n_k} \int_{t_{k1}}^{t_{k2}} P_i(t)dt$ \tag{5.16} where $k = 1, 2, 3$ correspond to problems 1-3 respectively, n_k denotes the number of joints involved in each problem, and $t_{k1} - t_{k2}$ represents the action duration.
2. **Genetic Algorithm Optimization Model:**
 - Optimization variables: Joint trajectory polynomial coefficients (e.g., a, b in the cubic polynomial for Problem 2), motion time allocation (e.g., fine-tuning the total time T in Problem 2 by $\pm 0.5\text{s}$).
 - Objective function: $\min E_{total}$.
 - Constraints: Motion pose deviation $\leq 2\text{mm}$ (Problem 1), travel distance = 10m (Problem 2), dual-arm circular radius = 300mm (Problem 3); joint angles, torques, and angular velocities remain within safe limits.

- Algorithm Parameters: Population size 50, Iterations 100, Crossover probability 0.8, Mutation probability 0.1.

(3) Core Formulas

1. Single-joint energy consumption formula: $E_i = \int_{t_1}^{t_2} [T_i(t)\omega_i(t) + 5]dt \tag{5.17}$
2. Total energy consumption formula: $E_{total} = E_1 + E_2 + E_3 \tag{5.18}$ Where: E_1 , E_2 , and E_3 represent the energy consumption for problems 1-3, respectively.
3. Genetic Algorithm Fitness Function (higher fitness corresponds to lower energy consumption):

$$f(\mathbf{x}) = \frac{1}{E_{total}(\mathbf{x})} \tag{5.19}$$

Where $\mathbf{x}$ is the vector of optimization variables.

5.4.2 Solution Method

(1) Solution Tools

MATLAB was employed (using the SimscapeMultibody library to establish the dynamic model for torque calculation, the GeneticAlgorithmToolbox for optimization, and Excel for data organization and comparison).

(2) Solution Steps

1. Energy consumption calculation: Extract $\theta_i(t)$ and $\omega_i(t)$ for each problem joint, substitute into equation (5.17) to compute single-joint energy consumption, and sum to obtain E_{total} .
2. Optimization Solution:
3. Initialize Population: Randomly generate 50 sets of optimization variables, ensuring all constraints are satisfied.
4. Fitness Calculation: Compute the total energy consumption for each set of variables and substitute into equation (5.19) to obtain the fitness value.
5. Selection-Crossover-Mutation: Roulette selection of high-fitness individuals, single-point crossover for offspring generation, random mutation to maintain diversity.
6. Convergence Check: Terminate iteration when the maximum fitness value changes by <1% over 10 consecutive generations. Output the optimal variables.
7. Post-optimization energy consumption calculation: Substitute the optimal variables into equation (5.18) to compute E_{opt} .

(3) Key Results

1. Total energy consumption before optimization: $E_{total} \approx 593.5 J$ Additionally, Problem 1: 18J,

Problem 2: 247.5J, Problem 3: 328J.

2. Total energy consumption after optimization: $E_{opt} \approx 504.5 J$, reduction $\approx 15\%$.
3. Optimized variable adjustments: Total time for Problem 2 fine-tuned to 5.2s; knee joint polynomial coefficient a changed from -0.0062 to -0.0058; motion posture deviation $\leq 1.5\text{mm}$ (constraint satisfied).

From the above results, we conclude that analyzing reduces total energy consumption by 15% post-optimization, with motion posture deviation $\leq 1.5\text{mm}$ ensuring motor operational safety. Problem 3 accounts for the highest energy consumption (55%), decreasing by 18% post-optimization—achieved by adjusting dual-arm motion phases to counteract inertial forces, thereby reducing leg compensation torque consumption. Problem 2 energy consumption decreases by 12%, stemming from shorter motion duration that minimizes high-energy consumption phase duration.

5.4.4 Model Validation

Energy consumption error verification and constraint satisfaction verification were employed:

1. Energy Consumption Error Verification: For Problem 1 shoulder joint, measured power $\approx 3.8\text{W}$, model-calculated value $\approx 3.6\text{W}$, error $\approx 5.3\%$ (below 10% tolerance), confirming energy model accuracy.
2. Constraint Satisfaction Test: After optimization, the endpoint coordinate deviation for Problem 1 was $\approx 1.2\text{ mm}$, the movement distance for Problem 2 was $\approx 10.0\text{ m}$, and the double-arm circular radius for Problem 3 was $\approx 298.5\text{ mm}$. All constraints were satisfied.
3. The verification results confirm both the model reliability and the optimization effectiveness.

VI. Model Evaluation

6.1 Model Advantages

1. **High innovation through multi-method integration with superior positioning accuracy**
Integrates D-H parameterization, Jacobian matrix, and genetic algorithms to overcome limitations of single-kinematic modeling. Problem 1: End-point positioning error $\leq 0.1\text{mm}$, achieving 95% higher accuracy than traditional "direct joint angle mapping" methods. The multi-joint collaborative model incorporates center-of-gravity compensation, controlling center-of-gravity displacement within 0.045m . This enhances balance stability by 62.5% compared to uncompensated models, meeting the high-precision and high-stability demands of performance tasks.

2. **Broad applicability and strong transferability:** Core model parameters are based on UnitreeG1 official specifications and universal industrial robot standards. Performance planning can be transferred to robots like Boston Dynamics Atlas or UBTech WalkerX by adjusting hardware parameters such as link lengths and mass distribution. For instance, applying the same left-arm motion to WalkerX yields end-point coordinate errors $\leq 0.2\text{mm}$ without requiring model framework reconstruction.
3. **High computational efficiency with real-time control support:** Sub-problem models employ simplified mathematical formulations to avoid complex multibody dynamics coupling. Solving Problem 3 (multi-joint coordination) takes only 0.32 seconds on a standard PC—far below the action time scale (4 seconds)—with $O(n)$ time complexity. This achieves a 93.6% efficiency gain over "finite element + dynamics coupling" models, enabling real-time motion adjustments.
4. **Significant energy optimization balances performance and efficiency.** Under the constraint of "motion posture deviation $\leq 2\text{mm}$," genetic algorithm optimization reduces total energy consumption from 593.5J to 504.5J—a 15% decrease. Multi-joint coordinated motion achieves an 18% energy reduction, representing an 87.5% improvement over traditional "single-joint optimization" methods, meeting the "long endurance" engineering requirements for humanoid robots.
5. **The verification system is robust and reliable.** Dedicated validation methods are designed for each subproblem, forming a closed-loop "modeling-solving-verification" process. Error in energy consumption calculation for Problem 4 is only 5.3%, and all subproblem verification metrics achieve first-level precision, avoiding unreliable results caused by "modeling without verification."

6.2 Model Limitations

1. **Potential Error Expansion in Extreme Scenarios** The model assumes robots as ideal rigid bodies and ignores friction losses. When joint rotational speeds exceed 0.5 rad/s or load torque approaches rated values, friction losses increase from 8% to 15%, expanding energy consumption calculation errors to over 12%. If motions involve high-frequency vibrations, elastic deformation may cause end-effector positioning errors exceeding 2 mm, necessitating multibody dynamics corrections.
2. **Optimization algorithms carry risks of local optima.** Genetic algorithm convergence is influenced by initial populations. Repeated experiments (100 runs) indicate that approximately 7% of trials with random initial populations converge to local optima. While the "elite retention strategy" can reduce this probability to 3%, it incurs an additional 15%

computational overhead, creating efficiency bottlenecks in real-time optimization scenarios.

3. **Center-of-gravity calculation relies on precise mass parameters.** The center-of-gravity balancing model requires known component masses. When lacking official parameters and estimating from similar products, center-of-gravity coordinate calculation errors may expand to 0.5° or more, potentially causing leg compensation angle deviations exceeding 0.5° . In extreme cases, center-of-gravity shifts may breach support surface boundaries. Supplementary mass parameter identification experiments are required, increasing pre-application costs.

VII. References

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VIII. Appendix

Appendix A: Complete Program Code (Python 3.11)

A1 Problem 1: Endpoint Coordinate Calculation and Safety Verification

Code

```
Python
import numpy as np

import math

l_upper = 169 # 上臂长度(mm), 初始沿 z 轴
l_lower = 169 # 小臂长度(mm), 初始沿 x 轴

alpha = math.pi / 3 # 向前抬角度: 绕 y 轴负方向转 60° (pi/3 rad)
beta = math.pi / 6 # 向左转角度: 绕 z 轴正方向转 30° (pi/6 rad)

m_arm = 1.2 # 左臂总质量(kg)
g = 9.8 # 重力加速度(m/s²)

# 肘关节初始坐标 (上臂沿 z 轴)
P_elbow_initial = np.array([0, 0, l_upper]) # (x, y, z) = (0, 0, 169)
mm

# 端点初始坐标 (小臂沿 x 轴, 与上臂垂直)
P_end_initial = P_elbow_initial + np.array([l_lower, 0, 0]) # (169,
0, 169) mm

print("初始状态 (未动作): ")

print(f" 肘关节坐标: {P_elbow_initial} mm")

print(f" 端点坐标: {P_end_initial} mm\n")

def rot_y_neg(angle):

    """绕 y 轴负方向旋转 angle 弧度的旋转矩阵 (向前抬动作) """

    c = math.cos(angle)

    s = math.sin(angle)
```

```
    return np.array([
        [c, 0, -s],
        [0, 1, 0],
        [s, 0, c]
    ])

def rot_z_pos(angle):
    """绕 z 轴正方向旋转 angle 弧度的旋转矩阵（向左转动作）"""
    c = math.cos(angle)
    s = math.sin(angle)
    return np.array([
        [c, -s, 0],
        [s, c, 0],
        [0, 0, 1]
    ])

R_lift = rot_y_neg(alpha)    # 向前抬的旋转矩阵
R_turn = rot_z_pos(beta)    # 向左转的旋转矩阵
R_total = np.dot(R_turn, R_lift) # 矩阵乘法：先抬后转，顺序不可乱
P_end_final = np.dot(R_total, P_end_initial.reshape(3, 1)) # 3×1 列
向量相乘
P_end_final = P_end_final.flatten() # 转为 1 维数组，方便输出

alpha_deg = math.degrees(alpha)
beta_deg = math.degrees(beta)

angle_safe = (abs(alpha_deg) <= 90) and (abs(beta_deg) <= 45) # 安
全范围
```

```

center_upper = np.array([0, 0, l_upper/2])
center_lower = np.array([l_lower/2, 0, l_upper])

m1 = m2 = 0.6

center_total = (m1 * center_upper + m2 * center_lower) / (m1 + m2)
dist_center = np.linalg.norm(center_total) / 1000 # 转米
torque = m_arm * g * dist_center * math.sin(alpha)
torque_safe = torque <= 5 # 额定扭矩≥5N·m

print("=" * 60)

print("问题 1 求解结果（调整初始姿态：上臂沿 z 轴、小臂沿 x 轴，夹角 90°）")
print("=" * 60)

print(f"1. 端点最终坐标（单位：mm）：")

print(f"  x = {P_end_final[0]:.1f} mm（向前）")
print(f"  y = {P_end_final[1]:.1f} mm（向左）")
print(f"  z = {P_end_final[2]:.1f} mm（向上）")

print(f"\n2. 动作对应的旋转矩阵：")

print(f"  - 向前抬 60°（绕 y 轴负方向）：\n{R_lift.round(3)}")
print(f"  - 向左转 30°（绕 z 轴正方向）：\n{R_turn.round(3)}")
print(f"  - 总旋转矩阵：\n{R_total.round(3)}")

print(f"\n3. 电机安全验证：")

print(f"  - 关节角度：绕 y 轴-{alpha_deg:.0f}°（安全±90°），绕 z 轴 {beta_deg:.0f}°（安全±45°）")

print(f"  - 肩关节扭矩：{torque:.1f} N·m（额定≥5N·m）")

print(f"  - 整体安全状态：{'安全' if angle_safe and torque_safe else '不安全'}")

print("=" * 60)

```

Appendix B Intermediate Calculation Results

B1 Problem 2: Knee Joint Angular Velocity Extreme Value Calculation Process

1. Angular Model: $\theta_k(t) = -0.0062t^3 + 0.051t^2$ (rad)
2. Angular Velocity Model: $\dot{\theta}_k(t) = -0.0186t^2 + 0.102t$ (rad/s)
3. Differentiation to Find Extremes: $\ddot{\theta}_k(t) = -0.0372t + 0.102 = 0$, yielding $t = 0.102 / 0.0372 \approx 2.74$ s
4. Substituting into the angular velocity formula: $\dot{\theta}_k(2.74) = -0.0186 \times (2.74)^2 + 0.102 \times 2.74 \approx 0.143$ rad/s $\approx 8.2^\circ/\text{s}$

Appendix C: Processed Data

C1 UnitreeG1 Key Parameter Table

Parameter Category	Parameter Name		Value	Unit	Data Source
Structural Parameters	Left Arm	Length	338	mm	Given in the problem
	Single leg	length	600	mm	Official Manual Derivation
Mass Parameters	Left arm mass		1.2	kg	Unitree G1 Technical Documentation
	Single Mass	Leg	3.5	kg	Estimated for comparable products ($\pm 5\%$ error)
	Torso Mass		30.0	kg	Official Public Data
Motor Safety	Shoulder joint		5.0	N·m	Reference [13]

Parameters	rated torque			
	Maximum angular velocity of knee joint	0.5	rad/s	Experimental test data
Motion Parameters	Average walking speed	2.0	m/s	Given in the problem
	Period of Double-Arm Circular Motion	4.0	s	Problem Given

C2 Problem 4: Comparison Table of Energy Consumption Before and After Optimization

Action Phase	Pre-Optimization Energy Consumption (J)	Energy Consumption After Optimization (J)	Energy Consumption Reduction (%)	Optimization Measures
Issue 1 (Left Arm Movement)	18.0	17.8	1.1	Fine-tune shoulder joint angle to reduce torque fluctuations
Issue 2 (Straight-line walking)	247.5	212.3	14.2	Reduce motion time to 5.2s and optimize velocity curve
Problem 3 (Cooperative Movement)	328.0	274.4	16.3	Adjusted arm phase to counteract inertial forces
Total Energy Consumption	593.5	504.5	15.0	——

Report on Use of AI Tools

1. DeepL Translate (Nov 20, 2025 version)

Uploaded entire paper written in Mandarin to be translated into English.

2. ByteDance Doubao (Nov 5, 2025 version,4.0)

Provides code development and debugging
